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RESEARCH ARTICLE OPEN ACCESS

Realistic Modeling of Photovoltaic Solar Cell: A Simple and Accurate Two-Diode Model

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Received: 29 October 2024 | **Revised:** 17 March 2025 | **Accepted:** 31 March 2025

Keywords: environmental conditions | Newton–Raphson method | solar cell | two-diode model

ABSTRACT

Photovoltaic modules are determinant in producing sustainable energy with a reduced environmental impact. This article explores the progressive modeling of photovoltaic modules, from the straightforward but approximate one-diode model to the more accurate but more complex two-diode model. It focuses on the parameters to be considered and the judicious choice of hypotheses to obtain electrical behavior close to that obtained experimentally for different environmental conditions. A reverse coupled saturation current and the Newton−Raphson method are both used for theoretical calculation and the simulation, respectively. Simulations show that the root mean square error (RMSE) on the I–V curves is reduced by 11.2% for irradiance of 1000 W/m^2 and by 28.3% on the P–V curves at 60°C. Additionally, the parallel resistance estimated with the two-diode model is lower than with the single-diode model (310 to 110.8 Ω), indicating a better consideration of leakage currents. Although the computation time is increased by around 40%, the improvement in accuracy justifies this added complexity. In conclusion, the study confirms the relevance of the two-diode model for a more realistic representation of photovoltaic module performance under various environmental conditions.

1 | Introduction

The consumption and production of energy are currently at the forefront of global concerns [1]. Global warming has prompted worldwide attention to the environmental impact of energy production. The use of fossil fuels is deemed unsustainable due to the ecosystem degradation it induces [2, 3]. The challenge lies in meeting the increasing energy demand while implementing highly efficient systems with very low or zero environmental impact. The scientific community has primarily focused on sustainable energy solutions to address this challenge [4]. Renewable energy sources such as hydroelectric, geothermal, wind, biomass, and photovoltaic (PV) energy have actually gained prominence [5]. Except for geothermal energy which puts to use the heat within the Earth, all other sources are derived directly or indirectly from solar energy, making them continuously available as long as there is sunlight. The PV technology is undergoing intensive research owing to its numerous advantages. One of the main challenges revolving around PV technology is to increase the efficiency, which, primarily, was supposed to be close to the Shockley−Queisser (SQ) limit, while simultaneously reducing the PV module cost [6, 7]. Indeed, a theoretical assessment of the efficiency limit of the PV single-junction led to a value of 33.9%. It is worth pointing out that the most important potential pathway to elucidate a reduction in PV performance is the thermalization energy loss

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of hot carriers. Thanks to different technologies developed around PV systems, these losses can be significantly reduced. By so doing, it would be possible to get closer to the SQ limit efficiency and give rise to the strong performance of the PV module. In this framework, hetero-junction cells can improve the efficiency of a solar installation. In a solar cell made with this technique, two different semiconductor materials, silicon and germanium, are generally involved. As these two materials have different electrical properties, the junction is more adapted to high operating temperatures. The PV panels using this technology are, therefore, more suited to hot climates. Indeed, the climatic range can go from −40°C to more than 90°C. The efficiency can be improved by overcoming negative consequences caused by hot carriers. This involves an in-depth study of the impacts of temperature on different physical quantities of the PV system, as well as the inevitable repercussions of these impacts on the performance of the considered PV module.

The evolutions of PV module parameters must be accurately described under in-service conditions to optimize their efficiency, which is greatly influenced by weather conditions (irradiance and temperature) [8, 9]. Analyzing PV module parameters is often done numerically to predict their behavior [10]. Accordingly, various analytical models have been developed for determining these parameters with different numerical strategies employed to estimate them from analytical solutions [11]. The one-diode and two-diode models are the most commonly used ones for PV module description. The complexity of the model depends on the number of parameters to be evaluated and on the environmental data. The single-diode model presents the advantage of simplicity because of a reduced number of considered parameters compared to the two-diode model. Generally speaking, five parameters are determined using the one-diode model. However, the single diode model has its limitations and also suffers from numerous approximations without accounting for all the physical phenomena involved in the operation of the PV module. The two-diode model offers multiple variants with an increased number of parameters considered (five to seven parameters), resulting in a more realistic PV module description [12]. This is due to the fact that some phenomena that take place during the PV module operation like electron-hole recombination and thermal loss are factored into this model.

The estimation of the parameters of the PV modules has been investigated theoretically and numerically in the literature considering the one-diode or two-diode models [13]. Numerous techniques to find adequate parameters have been proposed and evaluated in terms of ease of implementation and accuracy in determining the parameters of the PV modules and the calculation time [14–16]. Another challenge lies in developing an approach that accurately determines these parameters under realistic conditions, that is, taking into account the physics during the operation of the PV module and the influences of the environmental conditions. This should be done by balancing simplicity and computational time requirements [17]. Analytical, iterative, metaheuristic, or hybrid approaches are often used [13–22]. It should be noted that the one-diode model offers the advantage of benefiting from significant research developed using diverse approaches along with good accuracies [13]. The analytical methods are based on the use of a PV module data sheet or a current−voltage (I−V) characteristic curve for the determination of the parameters [18]. Regarding the iterative methods, at least one parameter is initialized to run the calculation of the parameters, followed by a repetitive process. Then, limitations are related to the approximation and the way that the initial value is set [19]. Metaheuristics enable to solve an optimization problem without any approximation considerations [20]. The hybrid methods combine two of the above methods or related algorithms. Several hybrid methods have been proposed with different complexities and efficiencies for estimating the parameters of the PV modules [14, 21, 22].

The estimation of the parameters of the PV module, initially approached with single-diode models, has evolved into more complex methods incorporating two-diode models and analytical, iterative, metaheuristic, or hybrid approaches. However, while these advancements enhance accuracy by accounting for real environmental conditions, they also require increasingly sophisticated algorithmic developments to balance precision and computational time. When studying a PV system with simulations involving an equivalent electrical circuit, the choice of the model has to be carefully argued. The present paper traces the progressive evolution in PV module modeling, transitioning from the single-diode to the two-diode model to improve the accuracy of the estimation and achieve this balance. The increasing complexity and number of parameters to be solved when shifting from the single-diode to the two-diode model is highlighted, with a particular focus on the selection of appropriate assumptions that enable faster convergence in numerical equation solving while achieving accuracy comparable to that of more elaborate methods. Successive improvements are demonstrated under different environmental conditions by comparing the performance of single and two-diode models, analyzing their impact on the accuracy of the estimated parameters, and evaluating their influence on the current-voltage (I−V) and power−voltage (P−V) characteristics of some PV modules.

The stages of the study are represented in the block diagram of Figure 1. After having mentioned the challenges of PV module modeling, a presentation of single and two-diode models and parameter estimation methods is given. An approach based on the two-diode model combined with the Newton−Raphson solving algorithm is then highlighted, whose performance is evaluated through comparisons with other models using numerical simulations and error analyses. Finally, the obtained results are discussed in terms of accuracy and computational time, leading to a conclusion on the relevance of the carefully chosen assumptions in the analytical formulation of the two-diode model.

2 | Theoretical Models of PV Modules

Depending on the theoretical approach adopted and the number of external and internal parameters considered, the calculated behaviors of the PV module that are found can be more or less realistic. The investigation carried out in this work is illustrated using the case of a silicon PV module with a maximum power of 49 W. The main objective is to obtain the value of the parallel resistance for which the PV module delivers

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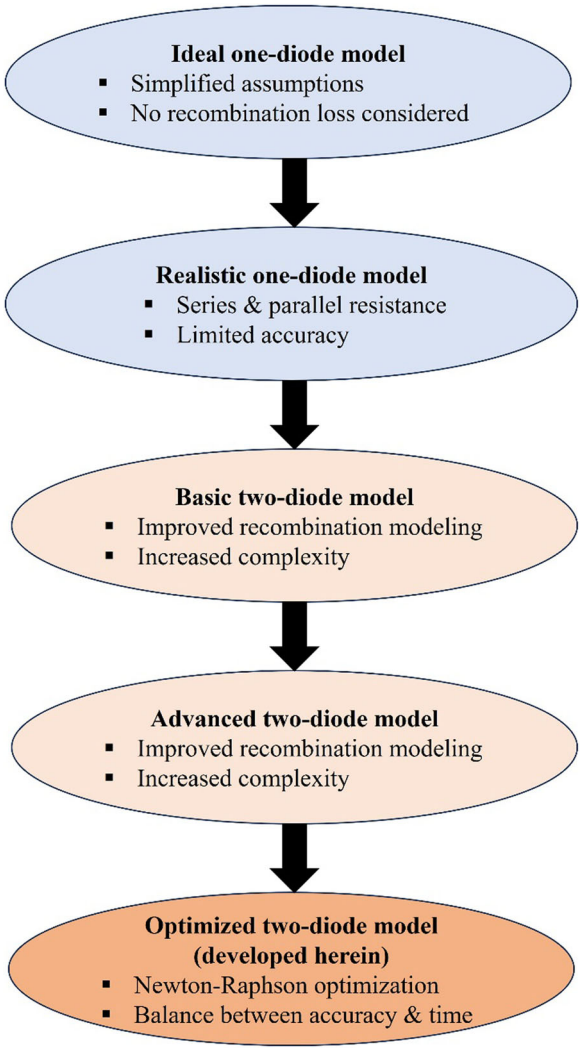

FIGURE 1 | Summary diagram of the article showing the evolution of the models and the main specific features.

continuously the maximum power of the installation. In that perspective, the I–V characteristics and the P–V of the PV module are determined. Three different configurations which depend on the number of the considered parameters are studied herein:

- An ideal model comprising one diode and one resistance in parallel or series;
- A comparable model with moderate complexity;
- A two-diode model.

2.1 | PV Model Using One Diode

The PV module can be likened to an electrical system for energy conversion. This model has been widely studied in the literature [23–25] thanks to its relative simplicity. This PV module is characterized by its equivalent electrical circuit which consists of a current source modeling the conversion of luminous flux into electrical energy, a shunt resistance, R_p as a consequence of the state of the material along the peripheric surface of the cell, a series resistance R_s representing the multiple resistances of

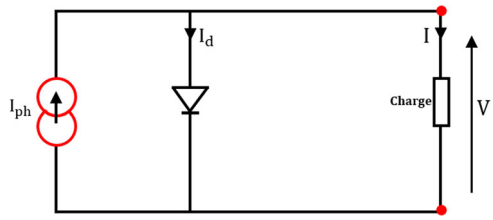

FIGURE 2 | Representation of an ideal one-diode photovoltaic model.

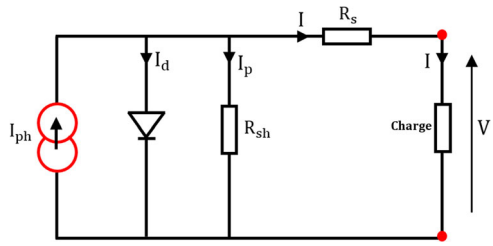

FIGURE 3 | Representation of a realistic one-diode photovoltaic model.

contact and connections, and a diode in parallel which models the PN junction (see Figure 2).

The single-diode model does not consider the effect of recombination current loss in the depletion region.

2.2 | PV Model Using Two Diodes

A realistic PV module takes into account different losses from external conditions and internal physical phenomena like electron-hole recombination events. These phenomena are considered by the insertion of a second diode. Recombination occurs when photogenerated electrons and holes recombine, then leading to the loss of photocurrent. The PV model with one diode does not consider the aforementioned internal phenomenon [26]. To implement a more realistic model of the PV module, the two-diode models can be considered [27–30]. In these configurations, an ideal solar cell is symbolized by a source of current associated with two diodes in parallel. The current source I_{ph} represents the photo-generated current. The branches of each of the diodes represent the diffusion current I_{d1} in the base and the emitter and the generation-recombination current I_{d2} in the space charge region of the junction.

In a practical situation, when a real solar cell is being considered, the following two resistances are added to the above components:

- A series resistance R_s, which models the resistive losses within the solar cell and therefore the metallization (see Figure 3);
- A parallel resistor R_p, modeling the leakage currents that short-circuit the junction (leaks from the edges of the cell and at the level of the junction).

In this context, the PV cell is assumed to be an equivalent electrical circuit as depicted in Figure 3.

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2.2.1 | Ideal Two-Diode Model: Five-Parameter Model

The equivalent circuit is obtained by simplifying the real model. To achieve this goal, it is assumed that the parallel resistance is infinite and the series resistance is equal to zero. It is worth noting that this model is only considered for theoretical exercises. It is, in fact, very far from the reality experienced on the ground. This model is depicted in Figure 4.

Here, I denote the total current produced by the cell. I_{ph} is the photo-generated current, I_{d_1} and I_{d_2} are the currents passing through the diodes d_1 and d_2, respectively. V is the voltage delivered by the cell. It should be noticed that V_{D_1} and V_{D_2} are the voltages imposed on the two diodes, respectively. By applying Kirchhoff's law, the output current of the cell takes the form:

$$I = I_{ph} - I_{d_1} - I_{d_2} \tag{1}$$

Within the framework of this model, we considered the two-exponential model [31, 32] for which the currents I_{d_1} and I_{d_2} crossing the two diodes are expressed as:

$$I_{D_1} = I_{0_1}\left[\exp\left(\frac{V_{D_1}}{A_1 \cdot N_S \cdot V_T}\right) - 1\right] \tag{2}$$

$$I_{D_2} = I_{0_2}\left[\exp\left(\frac{V_{D_2}}{A_2 \cdot N_S \cdot V_T}\right) - 1\right] \tag{3}$$

In these relations, $V_T = \frac{K \cdot T_C}{q}$ is the thermal voltage where K is the Boltzmann constant

$(= 138.10^{-23}\,J.K^{-1})$, q is the elementary charge of the electron $(= 16.10^{-19}\,C)$, and T_C is the cell temperature, and, I_{0_1} and I_{0_2} stand for the saturation currents of the two diodes d_1 and d_2, respectively. It can be assumed that:

$$I_{0_1} = I_{0_2} = I_0 \tag{4}$$

By applying the law of meshes:

$$V_{D_1} = V_{D_2} = V \tag{5}$$

The modified ideality factors of the two diodes, a_1 and a_2, are defined, respectively, by the following equations:

$$a_1 = A_1 \cdot N_S \cdot V_T \tag{6}$$

$$a_2 = A_2 \cdot N_S \cdot V_T \tag{7}$$

A_1 and A_2 are the ideality factors for the two diodes D_1 and D_2, respectively and N_S is the number of PV cells connected in series. To obtain the expressions (5) and (6), the thermal voltage V_T was considered, as defined in Equation (3). At 300K, $V_T = 26mV$ for the silicon cell.

Taking Equations (1)–(4), the following relationship among the parameters of the system is obtained:

$$I = I_{Ph} - I_0\left[\exp\left(\frac{V}{a_1}\right) + \exp\left(\frac{V}{a_2}\right) - 2\right] \tag{8}$$

This model is known as a five-parameter model as initially five parameters have to be determined, namely, I_{ph}, I_{0_1}, I_{0_2}, A_1, and A_2.

2.2.2 | Two-Diodes R_s-Model: Six-Parameter Model

In some configurations, the parallel resistance is assumed to be relatively very large. In such cases, the cell has only a series resistance. To approach reasonably the physical reality of a PV module, the following physical phenomena have to be considered in the design of the model:

- The losses induced by the contact resistance between the silicon and the surfaces of the electrodes;
- The natural resistance of the silicon material to the free flow of current and;
- The electrode resistance.

A series resistance R_s is introduced in the ideal model (see Figure 5) to improve the realistic description of the PV module. Furthermore, this model is based on six main parameters: I_{ph}, I_{0_1}, I_{0_2}, A_1, A_2, and R_s.

This model can be interpreted by the mathematical description of an electrical circuit comprising the connection in parallel of two diodes with the saturation currents I_{0_1} and I_{0_2}, the ideality factors A_1 and A_2, the source of current producing a photocurrent I_{ph} which depends on the irradiance and the series resistance. As evidenced by Equation (9), in this configuration, the I–V relationship can now be described accordingly.

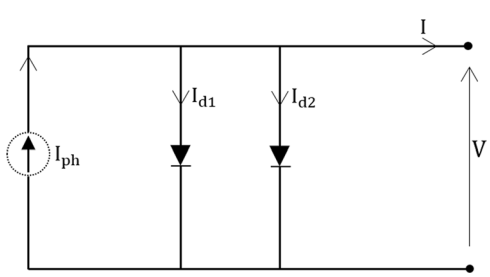

FIGURE 4 | Representation of an ideal two-diode photovoltaic model.

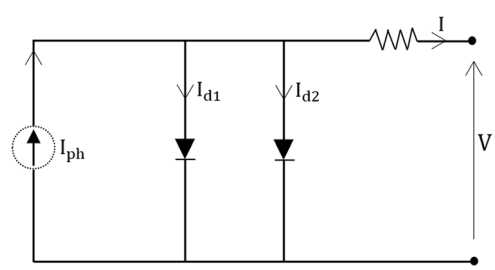

FIGURE 5 | Model of a PV cell with two diodes without parallel resistance.

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$$I = I_{ph} - I_0\left[\exp\left(\frac{V + R_S \cdot I}{a_1}\right) + \exp\left(\frac{V + R_S \cdot I}{a_2}\right) - 2\right] \quad (9)$$

To achieve this task, the law of nodes is written exactly as in the first case. In this model, the following expression for the saturation current common to the two diodes was used:

$$I_0 = I_{cc,ref}\exp\left(\frac{-V_{co,ref}}{a}\right)\left(\frac{T_c}{T_{c,ref}}\right)^3 \exp\left(\left(\frac{q \cdot \varepsilon_r}{\left(\frac{A_1 + A_2}{p}\right) \cdot K}\right)\left(\frac{1}{T_{c,ref}} - \frac{1}{T_c}\right)\right) \quad (10)$$

In this formula, the modified ideality factor, denoted by a, has been introduced:

$$a = \frac{N_s \cdot K \cdot T_c \cdot (A_1 + A_2)}{q \cdot p} \quad (11)$$

The normalization constant of the ideality factors, p, introduced in the two-diode model configuration will be determined later.

2.2.3 | Two-Diodes R_sR_p—Model: Seven-Parameter Model

The two-diode R_s–model presented above considerably degrades the quality of the expected results and is therefore unable to describe correctly the photocurrent production process when the silicon PV module faces strong temperature variations. In a realistic consideration of the PV module, one can record current leakage between the top gate and the back current. In this context, a parallel resistance R_p, also called short-circuit resistance, is incorporated to take this leakage into account. By so doing, the parameters of the model are increased from six to seven, namely: I_{ph}, I_{01}, I_{02} A_1, A_2, R_s, and R_p.

The final circuit of a realistic PV module representation is derived from the above two-diodes R_s-model, by adding a parallel resistance that symbolizes this form of current loss. This is shown in Figure 6. Taking into account R_P, the number of parameters evolves from six to seven.

By applying Kirchhoff's law, the current can be expressed as:

$$I = I_{ph} - I_{d1} - I_{shunt} - I_{d2} \quad (12)$$

with $I_p = I_{shunt}$.

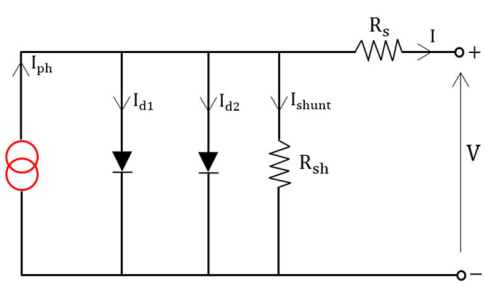

FIGURE 6 | Real model of a two-diode PV cell.

The application of the mesh law permits the formulation of this relation:

$$V_{D_1} = V_{D_2} = V_d = R_{sh} \cdot I_p \quad (13)$$

Then,

$$I_p = \frac{V_d}{R_{sh}} = \frac{V + R_s \cdot I}{R_{sh}} \quad (14)$$

where V_d is the common expression of the voltage applied to the terminals of each diode. Inserting Equations (2), (3), (4), and (14) into Equation (12), the expression of the output current becomes:

$$I = I_{ph} - I_0\left(\exp\left(\frac{q \cdot (R_s \cdot I + V)}{A_1 \cdot K \cdot T}\right) - 1\right) - I_0\left(\exp\left(\frac{q \cdot (R_s \cdot I + V)}{A_2 \cdot K \cdot T}\right) - 1\right) - \frac{V + R_s \cdot I}{R_{sh}} \quad (15)$$

Or by considering the modified ideality factors from Equations (6) and (7):

$$I = I_{ph} - I_0\left[\exp\left(\frac{V + R_S \cdot I}{a_1}\right) + \exp\left(\frac{V + R_S \cdot I}{a_2}\right) - 2\right] - \frac{V + R_S \cdot I}{R_{sh}} \quad (16)$$

2.2.4 | Comparison of the One-Diode and Two-Diode Models

The single-diode and two-diode models are the most used for PV module description. Each of these models presents advantages and drawbacks which are summarized in Table 1. The models are compared regarding the considered number of parameters, analytical formulation and numerical implementation complexities. The two-diode model because of its capacity to describe the PV parameters more realistically is employed here. This is due to the ability of this model to take into account the radiative phenomenon in PV modules during the operation by adding the second diode. The challenge herein is to propose a fairly straightforward estimation technique for the parameters. Accordingly, the Newton–Raphson method was considered to achieve the estimation of the parameters The Newton–Raphson method is known to be a fast and efficient algorithm for the simulation of the I-V curves and for the maximum power tracking, although its convergence is highly dependent on the choice of the initial approximation, the non-annulability of the derivative and the regularity of the function [33, 34]. Here, satisfactory accuracy and computation time were achieved by adjusting the set of initialized values and considering a criterion convergence of 10^{-6}.

3 | Numerical Model and Results

In its most realistic operational form, a two-diode PV module is very complex to manage, due to the very high number of unknowns to be accounted for. The digital process is time-computing [35].

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TABLE 1 | Comparison of the two models.

Consideration	Single-diode model	Two-diode model
Number of parameters	5	5–7
Equivalent circuit	Consists of a current source in parallel to a diode and an additional shunt resistance	Consists of a current source in parallel to two diodes and an additional shunt resistance
Physical consideration of phenomenon in PV module	No account of the recombination losses in the depletion region in the modeling	Non-radiative recombination is considered in the modeling process
	Relatively low accuracy especially at low illuminations	Good accuracy especially as irradiance gets lower
Analytical formulation	Can be accessed	More complex compared to the single-diode model
Numerical implementation	Less computation time	More time compared to the one-diode model when considering the same conditions
	The numerical implementation is relatively more accessible	The numerical implementation is more complex and elaborate

Our numerical approach consists of reducing the initial number of parameters so as to be able to propose solutions with reasonable calculation time. It is worth noting that the main equations governing the system are strongly nonlinear with double exponential terms. These parameters are also dependent on the influence of the incident solar radiation and the temperature of the PV device. Following the assumptions given by Ishaque et al. [27], the number of the parameters can be reduced by setting:

$$\frac{A_1 + A_2}{p} = 1 \tag{17}$$

By the Shockley diode diffusion theory, a value of $A_1 = 1$ can be set and a value of A_2 greater than 1.2 can be chosen. Using the standard test conditions (STC), the output current can be written as:

$$I = I_{Ph,ref} - I_{0,ref}\left[\exp\left(\frac{V}{a_{1,ref}}\right) + \exp\left(\frac{V}{a_{2,ref}}\right) - 2\right] \tag{18}$$

In the present work, $I_{Ph,ref}$ represents the photocurrent at the STC, $I_{0,ref}$ is the saturation current assigned at each single diode at STC. The modified ideality factors of the two diodes are represented by $a_{1,ref}$ and $a_{2,ref}$, respectively at the STC. At the point where the module is shortened ($I = I_{cc,ref}$ and $V = 0$), the inherent current deduced from Equation (14) is as:

$$I_{cc,ref} = I_{Ph,ref} - I_{0,ref}\left[\exp\left(\frac{0}{a_{1,ref}}\right) + \exp\left(\frac{0}{a_{2,ref}}\right) - 2\right] \tag{19}$$

which finally yields to:

$$I_{Ph,ref} \approx I_{cc,ref} \tag{20}$$

Two important environmental factors influence the photocurrent generation by the PV module, namely, the irradiance and the temperature. By rewriting this current according to these parameters, we obtain [34, 36]:

$$I_{Ph} = \frac{G}{G_{ref}}[I_{Ph,ref} + \mu_{sc}.\ \Delta T] \tag{21}$$

with $\Delta T = T_C - T_{C,ref}$.

Here, G stands for the irradiance. $G_{ref} = 1000W.\ m^{-2}$ is the irradiance at STC; $T_{C,ref} = 298K$ the module temperature at STC; $I_{Ph,ref}$, the intensity of the photocurrent at STC. μ_{sc} represents the temperature coefficient of the short-circuit current. This value is given by the manufacturer and is equal to $1.3 \times 10^{-3}K$ in our case.

Generally, it is assumed that the parallel resistance R_{sh} is relatively very large. Subsequently, the last term of the relation (16) can be neglected. Verifications were made according to the numerical values related to the PV modules under consideration. This equation of the output current has to be considered in three specific positions at STC:

- First, by introducing the voltage $V_{mp,ref}$ and the current $I_{mp,ref}$ at maximum power, the current/voltage relationship can be rewritten as:

$$I_{pm,ref} = I_{ph,ref} - I_{0,ref}\left[\exp\left(\frac{V_{pm,ref} + R_S \cdot I_{pm,ref}}{a_{1,ref}}\right) + \exp\left(\frac{V_{pm,ref} + R_S \cdot I_{pm,ref}}{a_{2,ref}}\right) - 2\right] \tag{22}$$

- Second, the voltage at the open circuit is determined. This is mathematically interpreted as:

$$I = 0 \text{ and } V = V_{oc,ref} \tag{23}$$

The substitution in Equation (16) yields:

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$$I_{ph,ref} = I_{0,ref}\left[\exp\left(\frac{V_{oc,ref}}{a_{1,ref}}\right) + \exp\left(\frac{V_{oc,ref}}{a_{2,ref}}\right) - 2\right] \quad (24)$$

where, $V_{oc,ref}$ is the voltage at the open circuit at STC and $I_{0,ref}$ is the saturation current at STC.

- Third, the current at the short circuit by setting is determined: $V = 0, I = I_{cc,ref}$ in the main Equation (16). The result is:

$$I_{cc,ref} = I_{ph,ref} - I_{0,ref}\left[\exp\left(\frac{R_S \cdot I_{CC,ref}}{a_{1,ref}}\right) + \exp\left(\frac{R_S \cdot I_{CC,ref}}{a_{2,ref}}\right) - 2\right] \quad (25)$$

Next, the basic Equation (16) is examined in a real physical configuration, both at STC and at the maximum power point. In such a context, R_s, R_{sh}, and the second diode are properly considered when modeling the system:

$$I_{mp,ref} = I_{ph,ref} - I_{0,ref}\left[\exp\left(\frac{V_{mp,ref} + R_s \cdot I_{mp,ref}}{a_1}\right) + \exp\left(\frac{V_{mp,ref} + R_s \cdot I_{mp,ref}}{a_2}\right) - 2\right] - \frac{V_{mp,ref} + R_s \cdot I_{mp,ref}}{R_P} \quad (26)$$

The approach considered herein consists in determining the values of the series and parallel resistances such that the maximum power determined numerically equals the maximum power determined experimentally. Then, Equation (27) is evidenced.

$$P_{mp,ref} = I_{mp,ref} \cdot V_{mp,ref} \quad (27)$$

Combining Equations (21) and (22) leads to the expression of the parallel resistance:

$$R_P = \frac{V_{mp,ref} + R_s \cdot I_{mp,ref}}{I_{ph,ref} - I_{0,ref}\left[\exp\left(\frac{V_{mp,ref} + R_s \cdot I_{mp,ref}}{a_1}\right) + \exp\left(\frac{V_{mp,ref} + R_s \cdot I_{mp,ref}}{a_2}\right) - 2\right] - \frac{P_{mp,exp}}{V_{mp,ref}}} \quad (28)$$

To obtain the expression of the saturation current at STC, Equation (21) is considered in the situation of an open circuit. Using Equation (20), Equation (29) is derived:

$$I_{0,ref} = \frac{I_{sc,ref}}{\exp\left(\frac{V_{oc,ref}}{a_1}\right) + \exp\left(\frac{V_{oc,ref}}{a_2}\right) - 2} \quad (29)$$

3.1 | Numerical Simulations

Each given value of R_s is associated by using computation with the corresponding parallel resistance (see Equation [23]) and the maximum power, P_{pm}. The process starts at $R_s = 0$. The value of R_s is gradually increased with the iteration until the absolute error between the maximum power computed and the experimental one provided by the manufacturer is minimized within a desired tolerance. Therefore, a suitable couple (R_P, P_{mp}) is found by the method of Newton–Raphson [37, 38].

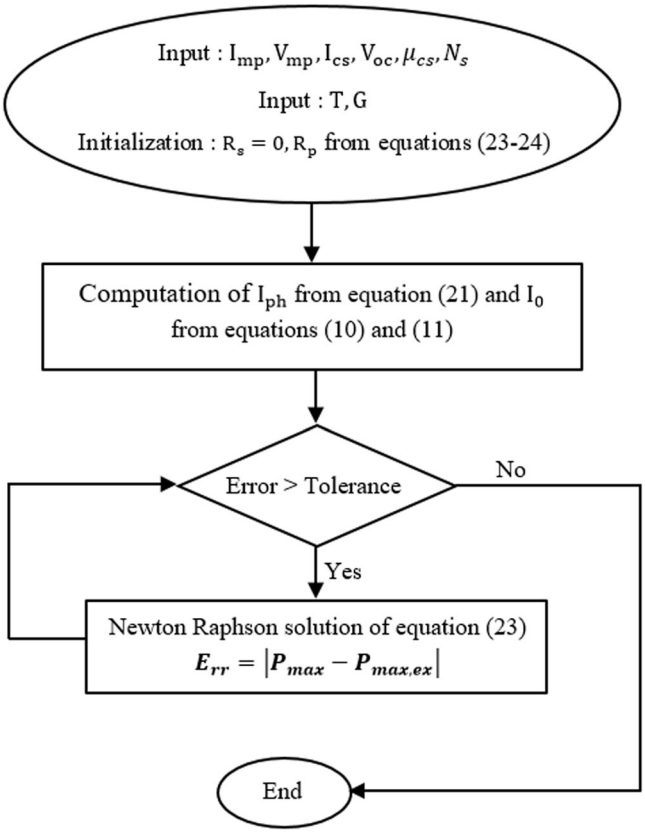

FIGURE 7 | Iteration flowchart.

To properly conduct our study and better assess the contribution of the second diode, the evolution curves of the main characteristics of the module as proposed in the work of Assia et al. [38] are superimposed on the same graph. The evolutions obtained in the configuration of the one-diode model in this former work and the ones in the configuration of the two-diode model that were developed herein were then compared. Calculations carried out enabled the representation of the characteristics I = f(V) and P = f(V) for the PV module under consideration.

To calculate the parameters of the model, the following sequence of steps is adopted (Figure 7):

1. The initial value of R_s is set to $R_s = 0$;
2. Then, the iterations are initiated: for the couple (R_s, R_{sh}) meticulously constructed, the corresponding power is evaluated;
3. The current value of the power is compared with the reference power given by the manufacturer. To achieve this, the validity test is carried out: $|P_m - P_{m,num}| < err$, with $err = 10^{-6}$). The current values of the couple (R_s, R_{sh}) are modified, and steps 2 and 3 are repeated until the validity criterion imposed on the power is verified.

At the end of the iterative procedure, the pair of resistances (R_s, R_{sh}) obtained is retained for subsequent operations.

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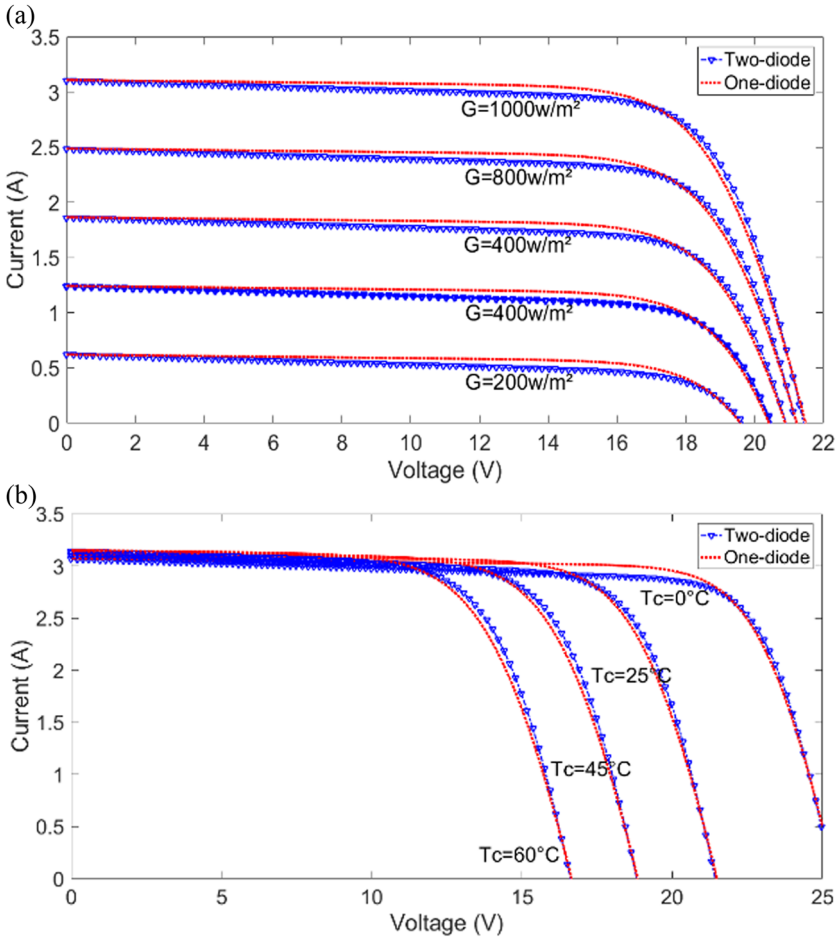

FIGURE 8 | Comparative analysis of I(V) for one-diode (curve in red) and two-diode (curve in blue) models for different irradiances (a) and temperatures (b).

3.2 | Results

Figure 8 represents the variation of the current as a function of the voltage for different values of the irradiance and the temperature, respectively. The results acquired with the one-diode model (in red) and the two-diode model (in blue) were superimposed. As expected, it is found that whatever the model, the maximum value of the current increases with irradiance (see Figure 8a). In addition, for a given value of the irradiance, the current reaches its maximum value at the origin of the voltage. This intensity is maintained as the voltage increases and falls suddenly when approaching the maximum voltage. The curves of the one and two-diode models are very close for low and intermediate voltages. The offset is detectable when the maximum voltage is approached. Also, it can be observed that the curve of the one-diode model is above the one of the two-diode model. It can therefore be concluded that in configuring the one-diode model, the maximal voltage was somewhat overestimated.

Figure 8b shows the variation of the current as a function of the voltage for different temperature values. The results obtained are as follows: the temperature has a weak impact on the maximum intensity for the two configurations, particularly for low voltages. The difference between the one-diode and the two-diode models is more noticeable when approaching high voltages. This shift can be attributed for all the cases to the failure to consider the recombination processes in the one-diode model. One can refer to the coefficient of ideality to propose some plausible explanations for the gap observed between the two curves at high voltages. Indeed, the ideality factor of a diode is a measure of how closely the diode follows the ideal diode equation. However, it has been demonstrated that at low voltages, the ideality factor of the second diode tends toward the value of 2 [39], while, at high voltages, the ideality factor tends toward 1, the ideal value. So, at high voltages, one can notice a significant improvement in the PV yield. This is a plausible explanation of the dropout of the curve of the two-diode model as compared to the one-diode model.

To estimate the quality of predictions, the root means square error (RMSE) was evaluated for different evolutions (Table 2). This quantity clearly indicates how far predictions fall from measured true values making use of Euclidean distance.

The RMSE is calculated from this relation:

$$E = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(A_i - B_i)^2} \tag{30}$$

Here, A_i designates the predicted value, B_i the observed value, and N, the number of observations.

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TABLE 2 | RMSE calculations of current evolutions with the irradiance and temperature.

G (W/m²)	T [°C]	E (I–V)	G (W/m²)	T [°C]	E (I–V)
200	25	0.1902	1000	0	0.0668
400	25	0.1963	1000	25	0.2142
600	25	0.2023	1000	45	0.4672
800	25	0.2082	1000	60	0.7405
1000	25	0.2142			

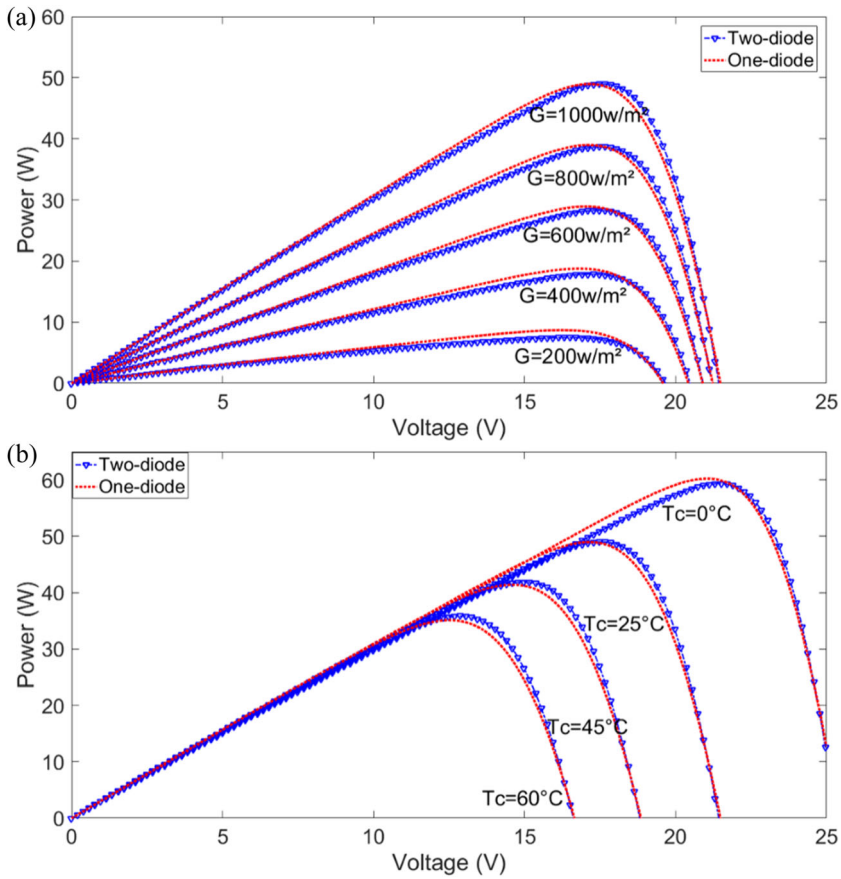

FIGURE 9 | Comparative analysis of P(V) for one-diode and two-diode models for different irradiances (a) and temperatures (b).

The results in Table 2 show that the calculated RMSE of the I–V evolutions increases slowly and gradually as the irradiance rises. The evaluated results deviate further from the values indicated by the manufacturer for high irradiances, although the variation is not very pronounced. The same conclusions are reached for the temperature as the RMSE increases with temperature, and also as the results tend to deviate from the manufacturer's values at high temperatures.

In Figure 9, the evolutions of the power as a function of the voltage for different values of the irradiance and the temperature are presented, respectively. It appears that the power increases with these two parameters in each case. It reaches the maximum value and suddenly decreases for high voltage values. The general trend is a power reduction when the second diode is integrated. The discrepancy that appears between these two curves always indicates that the one-diode model overestimates the power.

The RMSE can be also evaluated for the different curves. The results are summarized in Table 3. The increase of the solar irradiance is accompanied with the slight augmentation of the gap between the results of the two models used herein. Similar evolutions are obtained with the temperature, however, the gap with the temperature increase is more pronounced. This leads to the fact that the temperature variations result in phenomena that are not taken into account with the one-diode model.

From Table 3, it appears that the calculated RMSE of the power evolutions increases gradually with the irradiance. The observed shift does not vary too much with the irradiance. The same evolutionary trend of the RMSE calculations of power with the temperature is observed. However, the variation of the RMSE is much more marked with the increase in temperature. As an illustration, for a temperature of 25°C, the RMSE value is 5.0699, while for a relatively higher temperature of 60°C, the RMSE value goes up to 17.0755.

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TABLE 3 | RMSE calculations of power evolutions with the irradiance and temperature.

G (W/m²)	T [°C]	E (I–V)	G (W/m²)	T [°C]	E (I–V)
200	25	4.5084	1000	0	1.0779
400	25	4.6531	1000	25	5.0699
600	25	4.7945	1000	45	10.9785
800	25	4.9332	1000	60	17.0755
1000	25	5.0699			

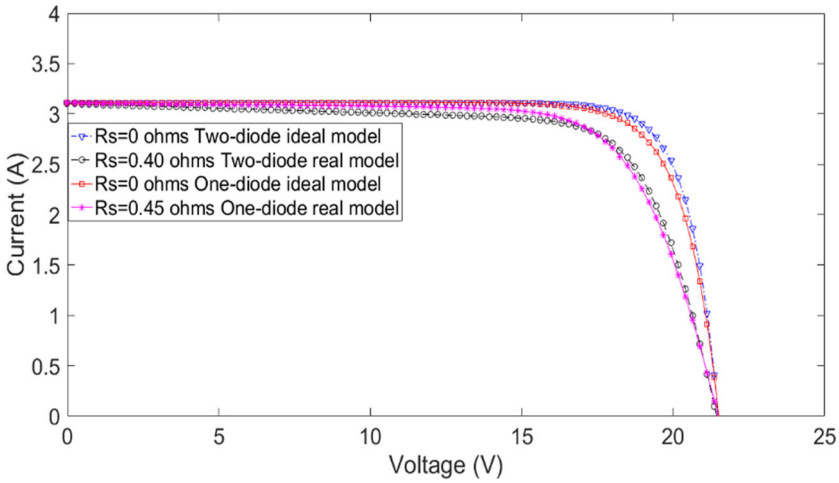

FIGURE 10 | Comparative analysis of I(V) for one-diode and two-diode models.

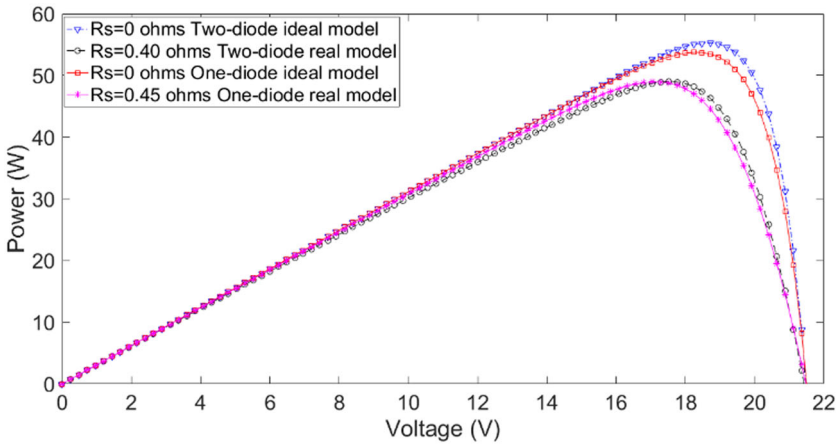

FIGURE 11 | Comparative analysis of P(V) for the one-diode and two-diode models.

In Figure 10, a comparative analysis of the impact of the series resistance on the I–V characteristic using the R_s model is made. The curves related to the two models have practically the same trend. At low voltages, they are almost superimposed. A slight offset is observed at high voltages. It can be justified by the fact that the recombination phenomenon is not considered for the one-diode model.

The evolution of the power in function of the voltage is shown in Figure 11. The curves have practically the same trend as in the case of irradiance (see Figure 9).

The main parameters of the PV module given by the manufacturer can be compared with the results of some models in the literature [25] and [29] and the one obtained by the two-diode model developed in this study. They are summarized in Table 4. These comparisons are made in STC

The maximum power is evaluated for different solar irradiance values and compared with the one-diode model results in [25] (see Table 5). It appears that the maximum power obtained in the context of a two-diode model is generally slightly lower than that obtained with the one-diode model for different values of the solar irradiance. This assertion is not verified except for an irradiance of 1000 W.m^{-2}. It can therefore be concluded that there was an overestimation of the maximum power in the one-diode model at low solar irradiances. Indeed, in this model, the leaks due to the recombination of the minority carriers are neglected.

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TABLE 4 | Comparison of the obtained results with the precedent studies [11, 12] in STC.

Manufacturer parameters		One-diode model [35]	Two-diode model [40]	Two-diode model developed herein
P_{mp}	49 W	49 W	49 W	49 W
I_{mp}	2.88 A			
V_{mp}	17 V			
I_0		2.9094×10^{-10}	1.782×10^{-10} 9.075×10^{-9}	2.9094×10^{-10}
I_p		0.0679	0.1049	0.1893
V_{OC}	21.8 V	21.47 V	21.47 V	21.47 V
$\mathbf{R_p}$	—	310.024 Ω	200 Ω	110.80 Ω
R_s	0.55 Ω	0.45 Ω	0.30 Ω	0.40 Ω
a		1.1172		0.9315
μ_{sc}	1.3×10^{-3}			
K_d	-72.5×10^{-3}			
N_s	36	36	36	36
a_1				0.9315
a_2				1.1178

TABLE 5 | Maximum power values as a function of the irradiance. The values shown in the table have been calculated for a cell temperature of Tc = 300 K.

Solar irradiance [W/m²]	Maximum power [W]	
	One-diode model [25]	Two-diode model (current work)
1000	48.95	49
800	39	38.97
600	28.92	28.82
400	18.77	18.6
200	8.685	8.395

TABLE 6 | Maximum power as a function of the temperature. Here, the irradiance is set at $G = 1000$ W. m^{-2}.

Temperature [°C]	Maximum power [W]	
	One-diode model [25]	Two-diode model (current work)
0	60.2	59.98
25	48.95	49
45	41.39	41.53
60	35.1	35.22

At this step of the work, the robustness of our model can also be evaluated by comparing its results to the one-diode model [25] for different temperature values. This approach enabled us to improve the accuracy and quality of the results. At the operating temperature of the module, the same maximum power proposed by the manufacturer, 49 W, is found (see Table 6). The curves representing the variation of the maximum power as a function of the irradiance and the temperature (the power is determined for different temperature values while irradiance is fixed), maintain the same profile for both the one-diode and the two-diode models.

The relative errors of each model with the manufacturer values were calculated as well (see Table 7). The obtained results were compared to those computed formerly with a single-diode model. A good agreement between the manufacturer parameters and the simulated data was observed. In the present work, the obtained results are consistent with those of the preceding work [25]. Indeed, it consists, in our case, in determining the value of the parallel resistance likely to generate the declared maximum power by the manufacturer. Our computational process led to the couple (R_s, R_p) = (0.40 Ω, 110.8 Ω).

With the single-diode model, the parallel resistance was estimated at 310 Ω [25]. The calculations run herein, as well as those of Assia et al. [40], considering the two-diode model led to a significant decrease in the parallel resistance value. In fact, the resistance dropped from 310 to 200 Ω in their work. However, our calculation gives a value of R_p = 110.80 Ω which is lower than the one obtained in [40]. It appears from this study that the current leakage was somewhat underestimated. The two-diode models result in the significant lowering of the parallel resistance, reflecting the greater current leakage, compared to that estimated in the one-diode model. In all the cases, the involvement of a second diode led to an increase in current losses (leakage or lost current). The consideration of electron-hole recombination, inevitable during the photocurrent generation, allowed us to get closer to reality and therefore improve the exactness of our results.

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TABLE 7 | Percentage of relative error of the obtained results and those of the one-diode model [25]. Calculations are carried out for Tc = 300 K and $G = 1000$ W. m^{-2}.

Reference parameter provided by the manufacturer PWX 500 PV (49 W)		One-diode model [25]	Two-diode model considered herein	% of relative error for the one-diode model [25]	% relative error of two-diode model developed herein
P_{mp}(W)	49	48.95	49	0.102	0
V_{oc}(V)	21.8	21.47	21.47	1.513	1.513
$R_S(\Omega)$	0.55	0.45	0.40	18.181	27.272
$R_P(\Omega)$	/	310.024	110.80	/	/
I_{SC}(A)	3.11	3.101	3.101	0.289	0.289

TABLE 8 | Comparison between the execution times of digital programs of the one-diode model [25] and the two-diode model developed herein.

The code	Code execution period of the one-diode model [25]	Code execution period of the two-diode model developed herein
The digital code analyzes the impact of temperature on the energy loss	0.0611 s	0.0869 s
The digital code analyzes the impact of irradiance on the energy loss	0.0669 s	0.0736 s
The digital code analyzes the energy loss as a function of the degradation rate of the series resistance	0.1805 s	0.2321 s

To go further in the evaluation of the capability of the model proposed herein, the calculation time is compared (see Table 8). It appears that the simulation time for the two-diode model is slightly higher than that of the one-diode model, although the difference is not significant. The additional time is attributed to the fact that in the two-diode model, the number of unknown parameters is raised from five to seven. However, the difference can be neglected.

4 | Conclusion

This article discusses the modeling of PV modules and aims to improve the accuracy of electrical parameter estimation by transitioning from the conventional single-diode model to the two-diode model. It emphasizes the assumptions that can be made to balance physical realism, estimation accuracy, and computational time in characterizing solar panel performance under various environmental conditions. While the single-diode model is simple and widely used, it has limitations, particularly in its inability to account for charge carrier recombination losses in the depletion region. To address these limitations, the two-diode model introduces a second diode, enabling a more accurate representation of internal phenomena such as non-radiative recombination and thermal losses. This transition increases the number of parameters to be estimated (from five to seven), making the calculations more complex but providing a better match with the actual performance of PV modules, especially under varying environmental conditions.

The calculations carried out here have shown behavior close to that found in the measured characteristics of a silicon PV module being considered. The simulation results clearly show that the performance of a PV module is strongly influenced by climatic conditions. The introduction of the second diode in the electrical equivalent circuit of the single-diode model increases the number of parameters to seven, adding the reverse saturation current I_0 and the ideality factor a_2 of the second diode. The determination of all the parameters of the model required specific theoretical calculations and numerical implementation such as the use of the Newton−Raphson method here. The values of the shunt and series resistances, R_p and R_s were chosen so that the numerical value of the power was equal to the experimental one provided in the manufacturer's datasheet under STC. The iteration process allowed the determination of the parallel resistance R_p. The two-diode model contributes significantly to decreasing the value of the parallel resistance R_p, from 310 Ω for a single-diode model to 110.8 Ω, indicating a better consideration of leakage currents. The discrepancies in errors are indicated by comparing successively more elaborate models. The comparative analysis shows that the two-diode model reduces the RMSE by 11.2% on I−V curves at 1000 W/m^2 and by 28.3% on P−V curves at 60°C compared to the single-diode model, at the cost of a 42% increase in computation time. The article highlights the impact of errors depending on the model chosen and the consequences that should be weighed up when simulating real conditions, particularly when seeking optimization of the performance of the PV systems by tracking the maximum power point, accounting for seasonal variations and quantifying the impact of aging.

Acknowledgments

The authors warmly appreciate the invaluable assistance of Mrs. Wilhelmina Logerais, a native English speaker, in revising and improving the linguistic quality of the paper.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are openly available in HAL at https://hal.science/.

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